

Selection Project

1. SELECTION DATA

Database: Level 2

Change...

Select from:

MaterialUniverse: Elastomers

2. SELECTION STAGES

Chart/Index

Limit

Tree

☒

Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

3. RESULTS: 8 OF 8 PASS

Show:

Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

Rank by:

Alphabetical

Name	X-Axis	Y-Axis
Butyl rubber (IIR)	910 - 950	7e-4 - 0,0015
Carbon black reinforced styrene...	1,13e3 - 1,15e3	0,0038 - 0,006
Ethylene vinyl acetate (EVA)	945 - 955	0,01 - 0,04
Natural rubber (NR)	930 - 970	0,0012 - 0,0021
Polychloroprene (Neoprene, CR)	1,23e3 - 1,3e3	0,00165 - 0,0...
Polyisoprene rubber (IIR)	930 - 970	0,0017 - 0,0026
Polyurethane (ePUR)	1,19e3 - 1,21e3	0,0025 - 0,03
Silicone elastomers (SI, Q)	1,02e3 - 1,22e3	0,005 - 0,05

Database: Level 2

Change...

Select from: MaterialUniverse: Composites

2. SELECTION STAGES

 Chart/Index  Limit  Tree

☒  Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

3. RESULTS: 5 OF 5 PASS

Show: Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

Rank by: Alphabetical

 Name	X-Axis	Y-Axis
 Aluminum/Silicon carbide comp...	2,66e3 - 2,9e3	81 - 100
 CFRP, epoxy matrix (isotropic)	1,5e3 - 1,6e3	69 - 150
 Dough (Bulk) molding compoun...	1,8e3 - 2,1e3	12 - 14
 GFRP, epoxy matrix (isotropic)	1,75e3 - 1,97e3	21 - 21,8
 Sheet molding compound, SMC,...	1,8e3 - 2e3	9 - 14

1. SELECTION DATA

Database: Level 2

Change...

Select from: MaterialUniverse: Foams

2. SELECTION STAGES

 Chart/Index  Limit  Tree

☒  Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

3. RESULTS: 8 OF 8 PASS

Show: Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

Rank by: Alphabetical

 Name	X-Axis	Y-Axis
 Ceramic foam	392 - 670	1,9 - 3,8
 Flexible Polymer Foam (LD)	38 - 70	0,001 - 0,003
 Flexible Polymer Foam (MD)	70 - 115	0,004 - 0,012
 Flexible Polymer Foam (VLD)	16 - 35	2,5e-4 - 0,001
 Metal foam	240 - 480	0,45 - 1,2
 Rigid Polymer Foam (HD)	170 - 470	0,2 - 0,48
 Rigid Polymer Foam (LD)	36 - 70	0,023 - 0,08
 Rigid Polymer Foam (MD)	78 - 165	0,08 - 0,2

Selection Project ✕

1. SELECTION DATA ▼

Database: Level 2 Change...

Select from: MaterialUniverse: Metals and alloys ▼

2. SELECTION STAGES ▼

Chart/Index
 Limit
 Tree

☒ Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

3. RESULTS: 28 OF 28 PASS ▼

Show: Stage 1: Young's modulus (GPa) vs. Density (kg/m³) ▼

Rank by: Alphabetical ▼

Name	X-Axis	Y-Axis	
Age-hardening wrought Al-alloys	2,67e3 - 2,84e3	68 - 76	
Brass	8,07e3 - 8,48e3	95 - 110	
Bronze	8,05e3 - 8,7e3	97 - 130	
Cast Al-alloys	2,65e3 - 2,77e3	69 - 76	
Cast iron, ductile (nodular)	7,05e3 - 7,15e3	170 - 180	
Cast iron, gray	6,94e3 - 7,23e3	94 - 140	
Cast magnesium alloys	1,75e3 - 1,87e3	42 - 47	
Commercially pure lead	1,13e4 - 1,14e4	13 - 14	
Commercially pure titanium	4,51e3	100 - 105	
Commercially pure zinc	7,13e3 - 7,15e3	90 - 110	
Copper	8,94e3	120 - 140	
Gold	1,93e4 - 1,94e4	76 - 81	
High carbon steel	7,8e3	200 - 220	
Lead alloys	1,05e4 - 1,13e4	14 - 17	
Low alloy steel	7,8e3	200 - 210	
Low carbon steel	7,8e3 - 7,82e3	200 - 220	
Medium carbon steel	7,8e3	200 - 220	
Nickel	8,83e3 - 8,95e3	190 - 220	
Nickel-based superalloys	7,88e3 - 8,68e3	200 - 220	
Nickel-chromium alloys	8,3e3 - 8,5e3	200 - 220	
Non age-hardening wrought Al-...	2,63e3 - 2,7e3	69 - 74	
Silver	1,05e4	69 - 73	
Stainless steel	7,61e3 - 7,87e3	190 - 210	
Tin	7,26e3 - 7,27e3	41 - 45	
Titanium alloys	4,43e3 - 4,79e3	110 - 120	
Tungsten alloys	1,69e4 - 1,86e4	310 - 370	

Selection Project

1. SELECTION DATA

Database: Level 2 Change...

Select from: MaterialUniverse: Polymers

2. SELECTION STAGES

Chart/Index
 Limit
 Tree

☒ Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

3. RESULTS: 21 OF 21 PASS			
Show:	Stage 1: Young's modulus (GPa) vs. Density (kg/m ³)		
Rank by:	Alphabetical		
Name	X-Axis	Y-Axis	
Acrylonitrile butadiene styrene (...)	1,03e3 - 1,06e3	2,07 - 2,76	
Cellulose polymers (CA)	980 - 1e3	1,66 - 1,74	
Epoxies (EP)	1,11e3 - 1,4e3	2,35 - 2,47	
Ionomer (I)	930 - 960	0,2 - 0,424	
Phenolics (PF)	1,24e3 - 1,32e3	2,76 - 4,83	
Polyamides (Nylons, PA)	1,12e3 - 1,15e3	0,94 - 2,04	
Polycarbonate (PC)	1,19e3 - 1,21e3	2,32 - 2,44	
Polyester (UP)	1,04e3 - 1,4e3	2,07 - 4,41	
Polyetheretherketone (PEEK)	1,3e3 - 1,32e3	3,76 - 3,95	
Polyethylene (PE)	939 - 960	0,621 - 0,896	
Polyethylene terephthalate (PET)	1,29e3 - 1,39e3	2,8 - 3	
Polyhydroxyalkanoates (PHA, P...	1,23e3 - 1,25e3	3,5 - 4	
Polylactide (PLA)	1,2e3 - 1,29e3	3,3 - 3,6	
Polymethyl methacrylate (Acryli...	1,17e3 - 1,2e3	2,24 - 3,24	
Polyoxymethylene (Acetal, POM)	1,39e3 - 1,41e3	2,6 - 3,2	
Polypropylene (PP)	895 - 909	0,824 - 1,02	
Polystyrene (PS)	1,04e3 - 1,05e3	3 - 3,5	
Polytetrafluoroethylene (Teflon,...	2,14e3 - 2,2e3	0,4 - 0,552	
Polyurethane (tpPUR)	1,12e3 - 1,24e3	1,31 - 2,07	
Polyvinylchloride (tpPVC)	1,29e3 - 1,46e3	2,2 - 3,1	
Starch-based thermoplastics (TPS)	1,2e3 - 1,5e3	1,32 - 1,34	

Selection Project

1. SELECTION DATA

Database: Level 2

Change...

Select from:

MaterialUniverse: Natural materials

2. SELECTION STAGES

Chart/Index

Limit

Tree

☒ Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

3. RESULTS: 9 OF 9 PASS

Show:

Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

Rank by:

Alphabetical

Name	X-Axis	Y-Axis
Bamboo	602 - 797	15,1 - 19,9
Cork	160 - 240	0,025 - 0,05
Hardwood: oak, across grain	850 - 1,03e3	5 - 5,58
Hardwood: oak, along grain	850 - 1,03e3	20,6 - 25,2
Leather	810 - 1,05e3	0,1 - 0,5
Paper and cardboard	700 - 1,15e3	2 - 4
Plywood	700 - 800	5 - 8
Softwood: pine, across grain	440 - 600	0,6 - 0,9
Softwood: pine, along grain	440 - 600	8,4 - 10,3

woods

Selection Project

1. SELECTION DATA

Database: Level 2

Change...

Select from:

MaterialUniverse: Technical ceramics

2. SELECTION STAGES

Chart/Index

Limit

Tree

☒

Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

3. RESULTS: 8 OF 8 PASS

Show:

Stage 1: Young's modulus (GPa) vs. Density (kg/m³)

Rank by:

Alphabetical

Name	X-Axis	Y-Axis
Alumina	3,66e3 - 3,74e3	322 - 338
Aluminum nitride	3,27e3 - 3,33e3	323 - 348
Boron carbide	2,49e3 - 2,55e3	362 - 380
Silicon	2,28e3 - 2,38e3	140 - 180
Silicon carbide	3e3 - 3,2e3	390 - 410
Silicon nitride	3,16e3 - 3,23e3	302 - 318
Tungsten carbides	1,53e4 - 1,59e4	600 - 670
Zirconia	5e3 - 5,8e3	195 - 205